# 鄭聖文 Sheng-Wen (Colin) Cheng

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## Summary

Roboticist and systems software engineer with 13+ years of hands-on experience building aerial robotic systems, embedded systems, and low-level software. Expertise spans flight control, state estimation, real-time systems, and hardware-software integration. Currently focused on applying optimal control and reinforcement learning to build intelligent autonomous systems.

## Employment

**NVIDIA** — Feb 2024 - Present
System Software Engineer — Taipei, Taiwan

- Develop Platform Security Controller (PSC) firmware to protect system integrity on Tegra-based platforms.
- Implement security features including boot image measurement, authentication, encryption, and decryption.

**Avilon Intelligence** — Sep 2018 - Mar 2021
Embedded Systems Engineer — Tainan, Taiwan

- Developed a carrier board for a 4G LTE system-on-module, enabling UAV remote control and video streaming.
- Built vision-based tag detection and localization software to enable autonomous drone landing.

## Education

**The University of Texas at Austin** — Aug 2025 - May 2027
M.S. in Computer Science — Texas, USA

- **Publication:** Published a paper at the American Control Conference (ACC 2026).
- **Presentation:** Speaker at Open Source Summit North America 2026, hosted by the Linux Foundation.

**National Taiwan University** — Sep 2022 - May 2024
Ph.D. Student in Electrical Engineering — Taipei, Taiwan

- **Presentation:** Speaker at Embedded Open Source Summit 2024, hosted by the Linux Foundation.

**National Yang Ming Chiao Tung University** — Sep 2019 - Nov 2021
M.S. in Robotics — Hsinchu, Taiwan

- **Master’s Thesis:** Design of Indoor-Outdoor Smooth Transferable Unmanned Aerial Vehicle
- Lead developer on a quadrotor research project conducted with the **Taiwan Space Agency (TASA)**

**Providence University** — Sep 2015 - Jun 2019
B.E. in Computer Science and Information Engineering — Taichung, Taiwan

- Graduated with honors; elected to the Phi Tau Phi Scholastic Honor Society (Taiwan).

## Projects

**NCRL Flight Control (Lead Developer)** [Video] [GitHub]

- Project licensed to the Taiwan Space Agency (TASA) for research on aggressive quadrotor maneuver control.

**semu: Minimalist RISC-V System Emulator (Core Developer)** [Video] [GitHub]

- Contributed to I/O virtualization of GPU, mouse, keyboard, storage, and random number generator via VirtIO.

**PU-01: Twin-boom Fixed-Wing UAV** [Video] [Slides]

- Determined airfoil aerodynamic parameters with XFLR5 stability analysis and fabricated the full airframe.

**Tenok: A Linux-like RTOS for Robotics and IoT [Video] [GitHub]**

- Built a POSIX-compliant RTOS optimized for ARM Cortex-M, adopting Linux-inspired kernel designs for robotics.

## Upstream Contributions

[1] **GNU C Library:** Fixed the RV32 THREAD_SELF register layout that affects GDB debugging [Patch]

[2] **Linux kernel:** Fixed a RISC-V KVM PMU counter-mask boundary issue [Patch]

## Publications

[1] **S.-W. Cheng** et al., "RotorBench: A Unified SE(3) Benchmark for Quadrotor Controllers under Disturbance and Model Uncertainty," *Journal manuscript in preparation.*

[2] **S.-W. Cheng** and T.-H. Cheng, "Data-Driven Estimation of Quadrotor Motor Efficiency via Residual Minimization," *American Control Conference (ACC)*, New Orleans, USA, 2026. [URL]

[3] **S.-W. Cheng** and Y.-H. Huang, "A Computationally Efficient GNSS/INS Design of Multirotor based on Error-state Kalman Filter," *62nd Annual Conference of the Society of Instrument and Control Engineers of Japan (SICE)*, Tsu, Japan, 2023. [URL]

[4] **S.-W. Cheng** and H.-A. Hung, "Robust State-Feedback $H_\infty$ Control of Quadrotor," *International Automatic Control Conference (CACS)*, Kaohsiung, Taiwan, 2022. [URL]

[5] S.-W. Wang, **S.-W. Cheng**, and C.-C. Huang, "Puyuma: Linux-based RTOS Experimental Platform for Constructing Self-Driving Miniature Vehicles," *Science and Information Conference (SAI)*, London, UK, 2018. [URL]

## Presentations

**International Conference Talks:**

[1] "Demystifying VirtIO-GPU: Building a Graphics Virtualization Bridge from Scratch," Open Source Summit North America (OSSNA 2026), Minneapolis, USA, 2026. [Link] [PDF]

[2] "Crafting a Vision-Aided Software Stack for UAV," Embedded Open Source Summit (EOSS 2024), Seattle, USA, 2024. [Link] [PDF]

**Keynotes & Invited Talks**

[1] "Trends in Machine Learning for Unmanned Aerial Vehicle Applications," Mobile Open Platform (MOPCON 2024), Keynote, Taiwan, 2024. [Link] [PDF]

[2] "Trends and lessons learned in deep learning and generative AI applications for UAV," PEGATRON Corporation, Taipei, Taiwan, 2024.

**Community Talks:**

[1] "The Illusion of Hardware: How Modern Virtual Machines Work," Conference for Open Source Coders, Users, and Promoters (COSCUP 2026), Taiwan, 2026. [Link] [PDF]

[2] "From Optimal Control to Reinforcement Learning for Quadrotor Drones," Sciwork Seminar 2026, Taiwan, 2026. [Link] [PDF]

[3] "Building a Quadrotor Simulator with Python: Modeling, Simulation, and Control," Open Tech Conference (OpenTechConf 2025), Hong Kong, 2025. [Link] [PDF]

[4] "Creating a Linux-like Real-Time Operating System for Quadrotor Drones," Conference for Open Source Coders, Users, and Promoters (COSCUP 2024), Taiwan, 2024. [Link] [PDF]

[5] "Tenok: Build a real-time operating system for Robotics," Conference for Open Source Coders, Users, and Promoters (COSCUP 2023), Taiwan, 2023. [Link] [PDF]